

SMART INDIA HACKATHON 2025



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HACKATHON
2025

Digital Krishi Officer

always available, always learning, always farmer first

Problem Statement ID: 25076

Problem Statement Title:

Ai-Based Farmer Query Support and Advisory System

Theme: Agriculture, Foodtech & Rural Development

PS Category: Software

Team ID:

Team Name: KrishiSetu



Digital Krishi Officer

How does it solve the problem?

Gives instant expert advice:

no waiting for officers

It has a local language support system:

removes literacy barrier

Gives personalized answers:

based on farmer's profile & region

It reduces Officer workload:

automates first-level queries

It's accessible anywhere:

works on mobile, even low bandwidth

It follows a learning Loop:

system gets smarter with every query



TECHNICAL APPROACH

Frontend: ReactJS + TailwindCSS + Rive

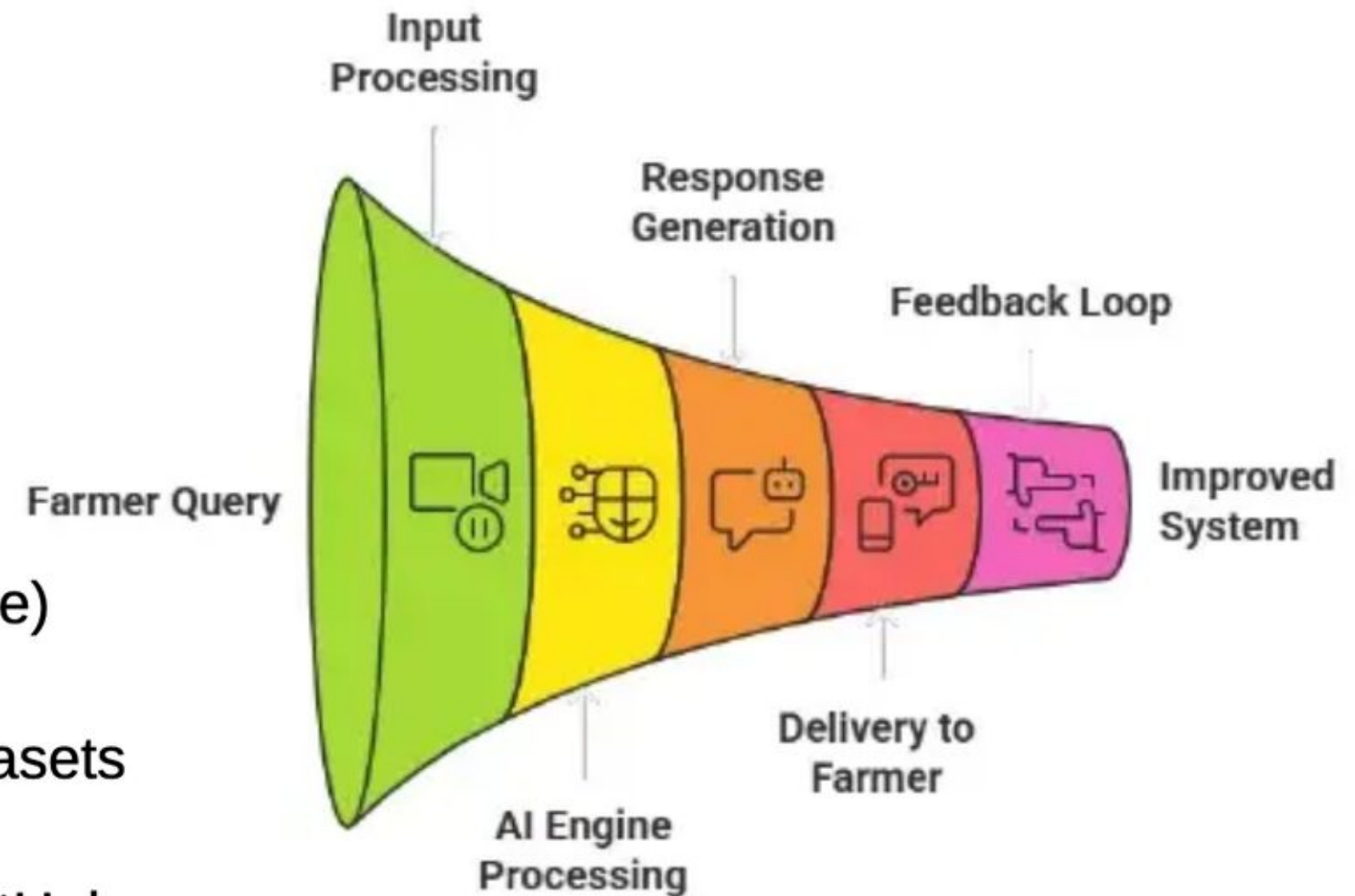
Backend: Node.js + Express

Database: MongoDB Atlas

Automation & Workflows: n8n (open-source)

AI Engine: Ollama (LLM) + Fine-tuned Agri Datasets

Other Tools: Cloud Storage (images/audio), GitHub (version control)



Feasibility

Technical: It uses open-source databases & is deployable on cloud.
Hence there is no high infrastructure cost.

Operational: It can work offline or using low connectivity networks
with local models & caching

Financial: It has a low-cost infrastructure (free-tier hosting, open-source
stack), easy to scale gradually

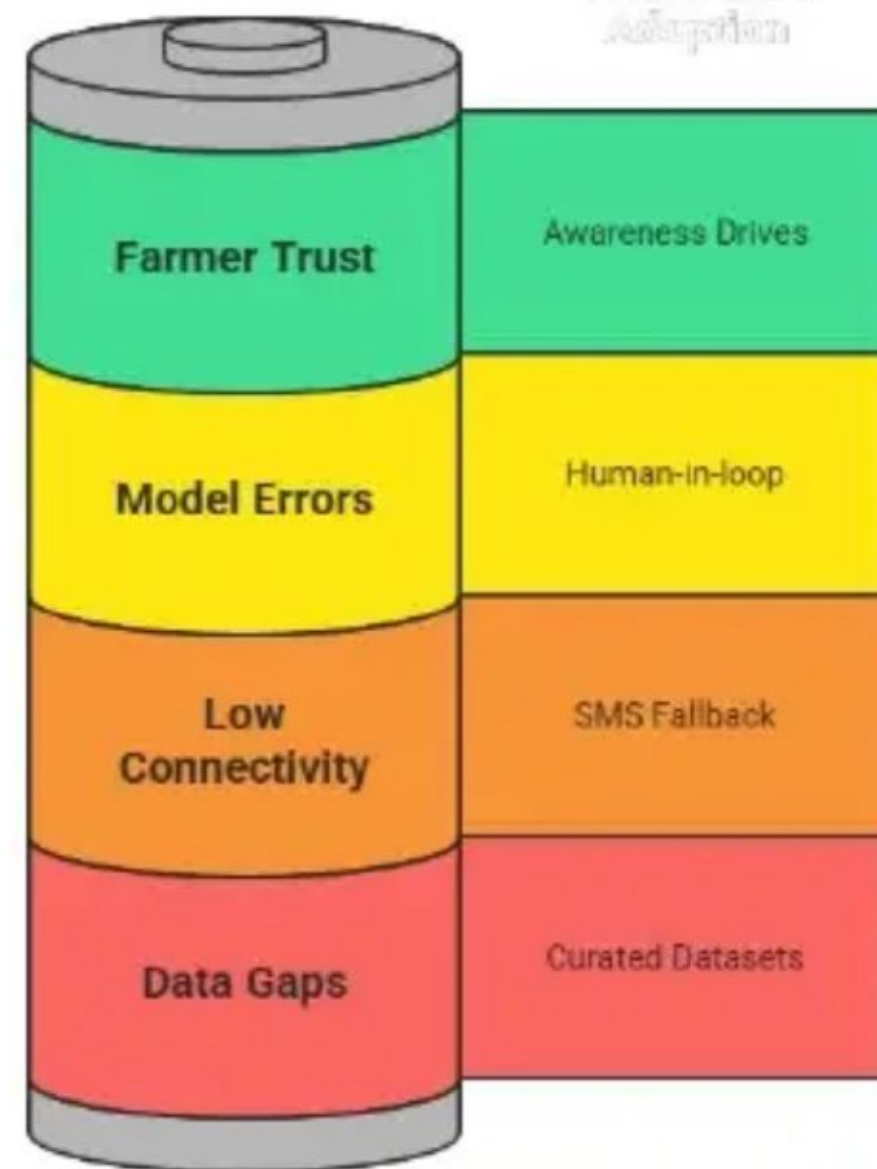
Challenges & Risks

Data Quality: Poor or missing agricultural datasets may reduce answer
accuracy

Connectivity: Farmers in remote areas may have poor internet access

Model Limitations: Hallucinations or wrong advice could mislead
farmers

Adoption: Some farmers may hesitate to trust AI



? Farmer Benefits

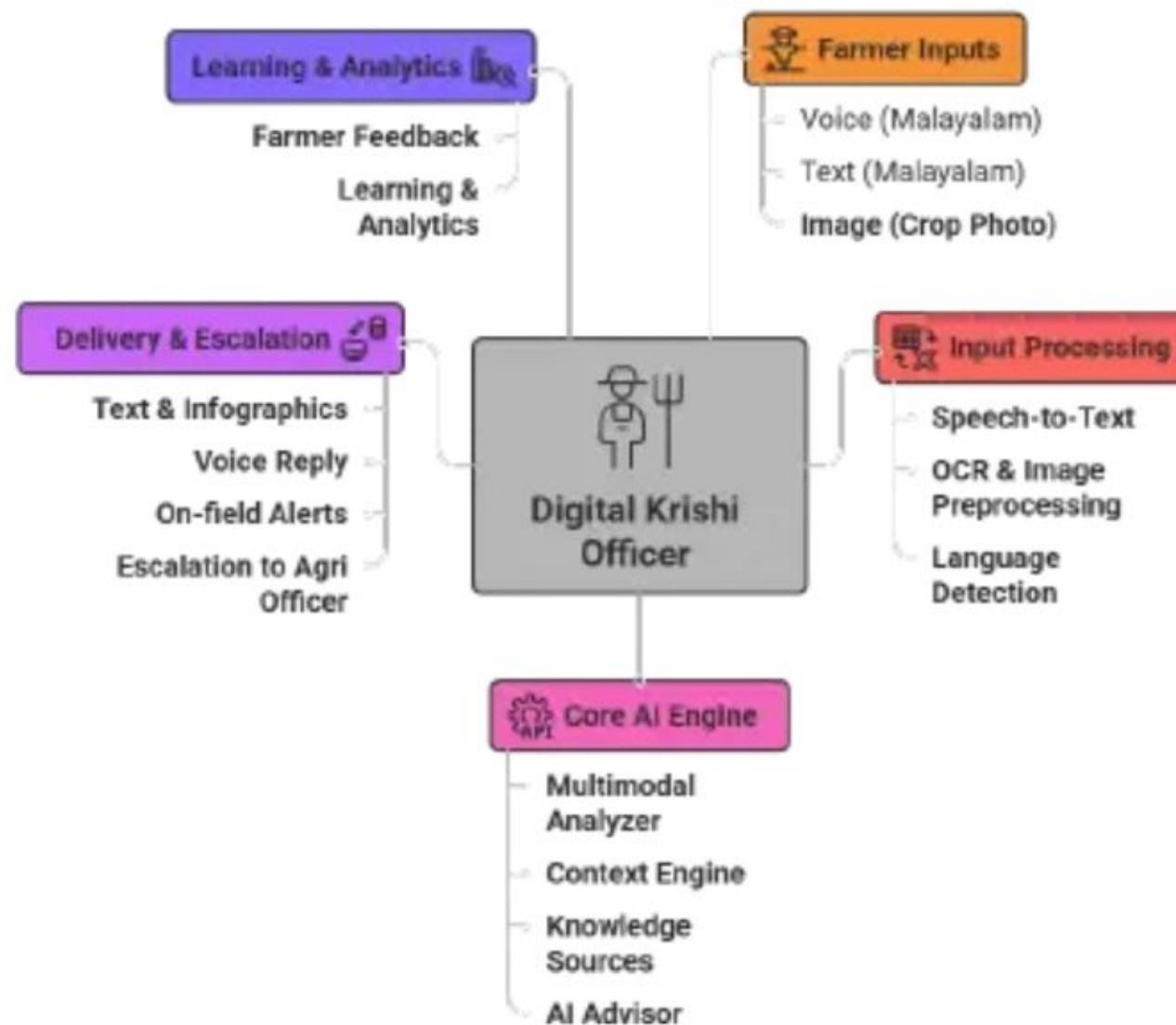
- Instant advice in Malayalam
- Better crop decisions
- higher yield
- Early pest/disease detection
- Voice & text support for all

? Community Impact

- Empowered, self-reliant farmers
- Higher productivity & lower losses
- Improved income & market access

Institutional Impact

- Reduces agri-officers workload.
- Scales to thousands of farmers
- Learns & improves continuously



🔍 Research & Reports

ICAR – Indian Council of Agricultural Research – Crop management & pest advisories

FAO Guidelines – Sustainable agriculture & integrated pest management

Kerala Agricultural University – Local crop calendars & soil recommendations

🔍 Online Resources

Government of India **Agri Portal** – Schemes, subsidies, and market trends

IMD (Indian Meteorological Department) – Weather & rainfall forecasts

eNAM – National Agriculture Market price data

🔍 AI & Tech References

Papers on **Multilingual NLP for Agriculture** (LLMs fine-tuned for Indic languages)

Research on **Image-based Plant Disease Detection** using CNNs

Case studies on **AI-based Rural Advisory Systems** (Digital Green, aWhere, Plantix)